PAU 2015

Criteris específics de correcció i qualificació per ser fets públics un cop finalitzades les proves **Tecnologia industrial**

SÈRIE 2

Primera part

Exercici 1

Q1 c **Q2** d **Q3** c **Q4** b **Q5** a

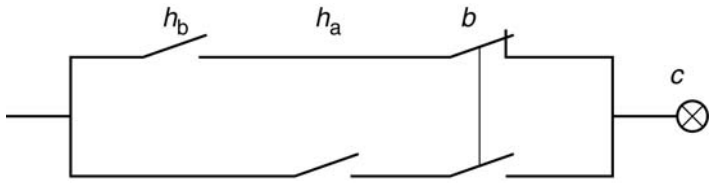
Exercici 2

a)

h_b	h_a	b	c
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	X ← No és possible
1	1	1	X ← No és possible

b) Amb X = 1 $c = h_b \cdot \bar{b} + h_a \cdot b$

c)



Segona part

OPCIÓ A

Exercici 3

a) $P_1 = P_{\text{elèctr}} \cdot \eta_{\text{mot}} = 1100 \cdot 0,76 = 836 \text{ W}$

b) $P_2 = P_1 \cdot \eta_{\text{transm}} \qquad n_2 = \tau \cdot n_1$

$$\Gamma_2 = \frac{P_2}{n_2 \frac{2\pi}{60}} = \frac{P_1 \cdot \eta_{\text{transm}}}{\tau \cdot n_1 \frac{2\pi}{60}} = 7,196 \text{ Nm}$$

c) $P_{\text{diss}} = P_{\text{elèctr}} - P_2 = P_{\text{elèctr}} \left(1 - \eta_{\text{mot}} \cdot \eta_{\text{transm}}\right) = 314,2 \text{ W}$

d) $\tau = \frac{n_2}{n_1} = \frac{d_1}{d_2} \quad \Rightarrow \quad d_2 = \frac{d_1}{\tau} = 112 \text{ mm}$

Exercici 4

a) $W = E_p = m\,g\,h = V\,\rho\,g\,h = 600 \cdot 1000 \cdot 9{,}807 \cdot 3{,}6 = 21{,}18\ \text{MJ}$

b) $P_h = \dfrac{W}{t} = 588{,}4\ \text{W}$

c) $\eta = \dfrac{W}{c \cdot \rho \cdot p_c} = 0{,}1955$

OPCIÓ B

Exercici 3

a) $E = P \cdot t \quad \Rightarrow \quad P = \dfrac{E}{t} = 2400\ \text{W}$

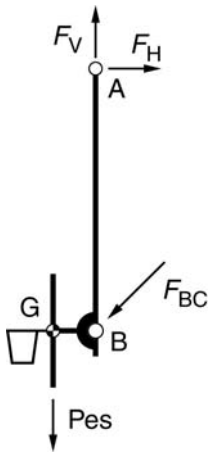
$I = \dfrac{P}{U} = 10{,}43\ \text{A}$

b) $R = \dfrac{U^2}{P} = \dfrac{\rho \cdot L}{A} \quad \Rightarrow \quad L = \dfrac{U^2 \cdot A}{P \cdot \rho} = 2{,}802\ \text{m}$

c) $S_{\grave{o}ptima} = 2{,}802 \cdot 1{,}5\,10^{-3}\ \text{m}^2 \quad \Rightarrow \quad S = \dfrac{100}{3{,}5} S_{\grave{o}ptima} = 0{,}12\,\text{m}^2$

Exercici 4

a)



b) $\sum M(\text{A}) = 0 \quad \rightarrow \quad m\,g\,L_2 - F_{BC} L_1 \cos\alpha = 0$

$$F_{BC} = \frac{m\,g\,L_2}{L_1 \cos\alpha} = \frac{35 \cdot 9{,}807 \cdot 0{,}5}{3 \cos 45^\circ} = 80{,}90\ \text{N}$$

c) $F_V - m\,g - F_{BC} \sin\alpha = 0 \quad \rightarrow \quad F_V = m\,g + F_{BC} \sin\alpha = 400{,}5\ \text{N}$

$F_H - F_{BC} \cos\alpha = 0 \quad \rightarrow \quad F_H = F_{BC} \cos\alpha = 57{,}21\text{N}$

d) $F_{\text{cable}} = F_{BC} \cos\alpha = 57{,}21\text{N}$